

Since $H_0^1(\Omega)$ is a Hilbert space with the norm $\|\cdot\|_{1,\Omega}$, the boundedness and coercivity assumptions on a , together with $F \in H^{-1}(\Omega)$, allow us to apply the Lax–Milgram lemma. Thus there is a unique $y \in H_0^1(\Omega)$ such that $a(y, v) = F(v)$ for every $v \in H_0^1(\Omega)$.

The finite-dimensional subspace V_h^0 is closed in $H_0^1(\Omega)$, so it is a Hilbert space with the inherited norm. The restrictions of a and F to this space retain the required boundedness and coercivity properties. A second application of the Lax–Milgram lemma therefore gives a unique $y_h \in V_h^0$ satisfying $a(y_h, v_h) = F(v_h)$ for every $v_h \in V_h^0$.

Let $e = y - y_h$. Subtracting the two variational equations for any $w_h \in V_h^0$ gives $a(e, w_h) = 0$. Hence, for any $v_h \in V_h^0$, the choice $w_h = y_h - v_h$ yields $a(e, e) = a(e, y - v_h)$. Coercivity and boundedness now imply $c\|e\|_{1,\Omega}^2 \leq a(e, e) \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}$. If $e \neq 0$, dividing by $c\|e\|_{1,\Omega}$ and taking the infimum over $v_h \in V_h^0$ proves $\|y - y_h\|_{1,\Omega} \leq (C/c) \inf_{v_h \in V_h^0} \|y - v_h\|_{1,\Omega}$. If $e = 0$, the same estimate is immediate.